

OTFS-ISAC Signal PAPR Suppression Based on Piecewise Companding

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Abstract—Orthogonal time-frequency space (OTFS) modulation is a promising candidate for integrated sensing and communication (ISAC) system in high-mobility scenarios. However, it suffers from high peak-to-average power ratio (PAPR), which will reduce power-amplifier efficiency and increases nonlinear distortion. To decrease the PAPR of OTFS-ISAC, a piecewise companding method is proposed in this paper. To ensure the communication performance, the signal-to-interference-plus-noise ratio (SINR) in communication receiver is also considered. First, the amplitude of the OTFS-ISAC signal is divided into five parts, and the relationship between the companding parameters and the SINR is derived by using the Bussgang theorem. Then, the optimization problem by minimizing the PAPR with SINR constraint is formulated, and the simultaneous perturbation stochastic approximation (SPSA) algorithm is applied to obtain the optimal companding parameters. Finally, simulation results demonstrate the effectiveness of the proposed method.

Index Terms—ISAC, OTFS, PAPR, Piecewise companding.

I. INTRODUCTION

Integrated sensing and communication (ISAC) has been widely regarded as a necessary technique in emerging 6G applications that demand high-speed communication and environment sensing, such as in intelligent transportation system [1]. Orthogonal frequency division multiplexing (OFDM) waveform has been widely adopted in ISAC systems due to its high spectral efficiency. However, in high-mobility scenarios, the Doppler shift destroys the orthogonality among subcarriers and leads to severe inter-carrier interference, which significantly deteriorates system performance.

Recently, Orthogonal time frequency space (OTFS) waveform has attracted significant attention in high-mobility scenarios. By modulating information within the delay-Doppler (DD)

domain, OTFS can effectively transform the rapidly time-varying fading channel into quasi-static channel [2], so that the performance of ISAC system can be improved. Nevertheless, OTFS suffers from a high peak-to-average power ratio (PAPR) due to multicarrier modulation [3]. High PAPR will reduce the power amplifier (PA) efficiency and make the PA work in the saturation region, which will cause the nonlinear loss of the OTFS. Therefore, the effective PAPR reduction method is essential for practical OTFS-ISAC systems.

The existing PAPR reduction techniques can be categorized into distortionless and distortion-based technique. The distortionless techniques will not cause the nonlinear distortion, such as probabilistic methods [4], [5] and coding methods [6], [7]. However, they usually have high computational complexity and require to transfer additional communication information, which will reduce the spectral efficiency. In contrast, the distortion-based technique has low computational complexity, for example the clipping technique and companding technique. The clipping technique introduces inevitable clipping noise, which degrades the bit error rate (BER) performance [8]. The companding technique compresses the signal dynamic range and make a favorable trade-off between signal degradation and PAPR reduction.

The companding technique includes nonlinear companding transforms (NCT) and linear companding transforms (LCT). The NCT can effectively decrease the PAPR of transmit signal, while it will introduce signal distortion and increase the BER [9], [10]. Compared with NCT, LCT usually has worse performance in terms of PAPR. However, it has better performance in terms of BER. For example, the three-stage linear companding (TSLC) method exploits the distribution of the envelope of the OTFS-ISAC signal, i.e., the Rayleigh distribution, and the segmentation thresholds are determined based on the cumulative distribution function (CDF) of the envelope of the OTFS-ISAC signal [11]. However, the companding coefficients in the TSLC method is fixed, and they are

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not suitable for different OTFS-ISAC signals, which limits the performance of the OTFS-ISAC signal in terms of PAPR and BER.

To improve the performance of the OTFS-ISAC signal, a piecewise companding method is proposed in this paper. In this method, the amplitude of OTFS-ISAC signal is divided into five regions based on the CDF and probability density function (PDF) of the OTFS-ISAC signal envelope. Then, the Bussgang theorem is employed to evaluate the companding distortion, and the PAPR is minimized under the signal-to-interference-plus-noise ratio (SINR) constraint. Finally, companding parameters are obtained by simultaneous perturbation stochastic approximation (SPSA) algorithm.

The residue of this paper is organized as follows. The signal model is given in Section II. The piecewise companding method is proposed in Section III. In Section IV, several simulation results are given. Finally, conclusions are drawn in Section V.

II. SIGNAL MODEL

A. OTFS-ISAC Signal Model

Consider the OTFS-ISAC signal with N Doppler bins and M delay bins. Let $\mathbf{X}_{DD} \in \mathbb{C}^{M \times N}$ denote the transmitted symbol matrix in the DD domain. The symbols are mapped to the time-frequency domain via the inverse symmetric finite Fourier transform (ISFFT) and then transformed to the time domain using the Heisenberg transform.

The vectorized time-domain transmit signal $\mathbf{x}_T \in \mathbb{C}^{MN \times 1}$ can be expressed as

$$\mathbf{x}_T = \text{vec}(\mathbf{F}_M^H (\mathbf{F}_M \mathbf{X}_{DD} \mathbf{F}_N^H)) = \text{vec}(\mathbf{X}_{DD} \mathbf{F}_N^H) = \mathbf{S} \mathbf{x}_{DD} \quad (1)$$

where $\mathbf{F}_M$ and $\mathbf{F}_N$ denote the normalized M -point and N -point discrete Fourier transform matrices, respectively. $\text{vec}(\cdot)$ denotes column vectorization, $\mathbf{x}_{DD} = \text{vec}(\mathbf{X}_{DD})$, and $\mathbf{S} = (\mathbf{F}_N^H \otimes \mathbf{I}_M)$, in which $\otimes$ represents the Kronecker product.

Let $x_q = |x_q| e^{j\varphi_{x_q}}$ denote the q -th element of $\mathbf{x}_T$, where $|x_q|$ and φ_{x_q} represent the magnitude and phase of the q -th element of $\mathbf{x}_T$, respectively. Suppose that the data symbols, i.e., the elements in $\mathbf{x}_{DD}$, of the OTFS-ISAC signal are approximately independent. When the number of symbols N and subcarriers M are sufficiently large, the real and imaginary parts of the time-domain OTFS-ISAC samples x_q can be approximated as Gaussian random variables. Consequently, the amplitude of the time-domain OTFS-ISAC sample $|x_q|$ follows a Rayleigh distribution, i.e.,

$$\mathcal{P}(|x_q|) = \frac{2|x_q|}{\sigma^2} \exp\left(-\frac{|x_q|^2}{\sigma^2}\right) \quad (2)$$

where σ^2 denotes the average power of the transmitted signal in the time domain. The CDF of $|x_q|$ is

$$\mathcal{F}(|x_q|) = 1 - \exp\left(-\frac{|x_q|^2}{\sigma^2}\right) \quad (3)$$

B. Piecewise Companding OTFS-ISAC Signal Model

According to (3), we can see that the OTFS-ISAC signal mainly possesses the medium amplitude, and the small and large amplitude samples are with low probability. Motivated by these statistical characteristics, the piecewise companding method is proposed to suppress the PAPR of the OTFS-ISAC signal. In this method, the amplitude of the OTFS-ISAC signal is divided into five parts by the amplitude mapping function.

The amplitude mapping function $\mathcal{M}(x)$ is determined by four thresholds $\{s_k\}_{k=1}^4$ and three compression factors $\{u_k\}_{k=1}^3$. These thresholds satisfy $0 < s_1 < s_2 < s_3 < s_4$. Moreover, s_2 and s_3 should satisfy $\mathcal{F}(s_2) = a_2$ and $\mathcal{F}(s_3) = a_3$, respectively. s_1 and s_4 should satisfy $\mathcal{P}(s_1) = b_1$ and $\mathcal{P}(s_4) = b_4$, respectively, where a_2 and a_3 denote the cumulative probabilities, and b_1 and b_4 represent the specific probability density values. $\mathcal{M}(|x_q|)$ can be expressed as

$$\mathcal{M}(|x_q|) = \begin{cases} (u_1 |x_q| + v_1) & |x_q| \leq s_1 \\ (u_2 |x_q| + v_2) & s_1 < |x_q| \leq s_2 \\ |x_q| & s_2 < |x_q| \leq s_3 \\ (u_3 |x_q| + v_3) & s_3 < |x_q| \leq s_4 \\ (u_3 s_4 + v_3) & s_4 < |x_q| \end{cases} \quad (4)$$

where v_1, v_2 , and v_3 are offsets to ensure the continuity of $\mathcal{M}(|x_q|)$.

Let $y_q = \mathcal{M}(|x_q|) e^{j\varphi_{x_q}}$ denote the piecewise companding signal of x_q , i.e., the q -th element of the companded signal $\mathbf{y}_T$. The PAPR of the companded signal $\mathbf{y}_T$ is

$$\text{PAPR} = \frac{\max_q |y_q|^2}{\frac{1}{MN} \sum_{q=0}^{MN-1} |y_q|^2} = \frac{MN \max_q |y_q|^2}{\sum_{q=0}^{MN-1} |y_q|^2} \quad (5)$$

III. PIECEWISE COMPANDING METHOD

A. Optimization Problem

In this section, the companding parameters are optimized to minimize PAPR under the SINR constraint. Under the complex Gaussian distribution assumption, the Bussgang theorem is used [12] to decompose the companded signal $\mathbf{y}_T$ as

$$\mathbf{y}_T = \alpha \mathbf{x}_T + \mathbf{d} \quad (6)$$

where α is the Bussgang gain and $\mathbf{d}$ denotes the nonlinear distortion. The Bussgang gain α is obtained by minimizing $\mathbb{E}[\|\mathbf{y}_T - \alpha \mathbf{x}_T\|_2^2]$, giving by

$$\alpha = \frac{\mathbb{E}[\mathbf{x}_T^H \mathbf{y}_T]}{\mathbb{E}[\mathbf{x}_T^H \mathbf{x}_T]} \approx \frac{\sum_{q=0}^{MN-1} x_q^* y_q}{\sum_{q=0}^{MN-1} |x_q|^2} \quad (7)$$

A key property of decomposition in (6) is that the distortion $\mathbf{d}$ is uncorrelated with the input $\mathbf{x}_T$, i.e., $\mathbb{E}[\mathbf{x}_T^H \mathbf{d}] = \mathbf{0}$. The corresponding distortion power σ_{dist}^2 is defined as

$$\sigma_{\text{dist}}^2 = \frac{1}{MN} \mathbb{E}[\|\mathbf{y}_T - \alpha \mathbf{x}_T\|_2^2] = \mathbb{E}[|y_q|^2] - |\alpha|^2 \mathbb{E}[|x_q|^2] \quad (8)$$

Hence, we further assume that the distortion $\mathbf{d}$ satisfies $\mathbb{E}[\mathbf{d} \mathbf{d}^H] \approx \sigma_{\text{dist}}^2 \mathbf{I}_{MN}$.

The received signal $\mathbf{r}_T \in \mathbb{C}^{MN \times 1}$ in the communication receiver can be represented as

$$\mathbf{r}_T = \mathbf{H}_c \mathbf{y}_T + \mathbf{n}_c = \alpha \mathbf{H}_c \mathbf{x}_T + \mathbf{H}_c \mathbf{d} + \mathbf{n}_c \quad (9)$$

where $\mathbf{H}_c \in \mathbb{C}^{MN \times MN}$ is the time-domain channel matrix, and $\mathbf{n}_c \sim \mathcal{CN}(\mathbf{0}, \sigma_c^2 \mathbf{I})$ is the additive white Gaussian noise (AWGN).

Applying the Wigner transform and the symplectic finite Fourier transform (SFFT) on $\mathbf{r}_T$ yields the DD-domain received vector $\mathbf{r}_{DD} \in \mathbb{C}^{MN \times 1}$ as

$$\mathbf{r}_{DD} = \alpha \mathbf{H}_{c,DD} \mathbf{x}_{DD} + \mathbf{S}^{-1} (\mathbf{H}_c \mathbf{d} + \mathbf{n}_c) \quad (10)$$

where $\mathbf{S}^{-1} = (\mathbf{F}_N \otimes \mathbf{I}_M)$ and $\mathbf{H}_{c,DD} = \mathbf{S}^{-1} \mathbf{H}_c \mathbf{S}$.

Assume the zero forcing equalizer $\mathbf{W}_{DD} = \mathbf{H}_{c,DD}^{-1}$ is used in the communication receiver. The equalized output is

$$\hat{\mathbf{r}}_{DD} = \mathbf{W}_{DD} \mathbf{r}_{DD} = \alpha \mathbf{x}_{DD} + \mathbf{W}_{DD} \mathbf{S}^{-1} (\mathbf{H}_c \mathbf{d} + \mathbf{n}_c) \quad (11)$$

Define the detection error vector as

$$\mathbf{v} = \frac{\hat{\mathbf{r}}_{DD}}{\alpha} - \mathbf{x}_{DD} = \frac{1}{\alpha} (\mathbf{S}^{-1} \mathbf{d} + \mathbf{W}_{DD} \mathbf{S}^{-1} \mathbf{n}_c). \quad (12)$$

Hence, the error covariance matrix $\mathbf{R}_v \in \mathbb{C}^{MN \times MN}$ is

$$\mathbf{R}_v = \mathbb{E}[\mathbf{v} \mathbf{v}^H] = \frac{\sigma_{\text{dist}}^2 \mathbf{I} + \sigma_c^2 \mathbf{W}_{DD} \mathbf{W}_{DD}^H}{|\alpha|^2} \quad (13)$$

The mean square error (MSE) of the n -th symbol is the n -th diagonal element of the error covariance matrix, i.e., $\text{MSE}_n = [\mathbf{R}_v]_{n,n}$. For unit-power symbol, the SINR is expressed as

$$\text{SINR}_n = \frac{1}{\text{MSE}_n} = \frac{|\alpha|^2}{\left[\sigma_{\text{dist}}^2 \mathbf{I} + \sigma_c^2 \left(\mathbf{H}_{c,DD}^H \mathbf{H}_{c,DD} \right)^{-1} \right]_{n,n}} \quad (14)$$

To minimize the PAPR of the OTFS-ISAC signal and ensure the BER, the following optimization problem is formulated as

$$\begin{aligned} & \min_{\mathbf{u}} \frac{MN \max_q |y_q|^2}{\sum_{q=0}^{MN-1} |y_q|^2} \\ & \text{s.t.} \begin{cases} \frac{|\alpha|^2}{\left[\sigma_{\text{dist}}^2 \mathbf{I} + \sigma_c^2 \left(\mathbf{H}_{c,DD}^H \mathbf{H}_{c,DD} \right)^{-1} \right]_{n,n}} \geq \gamma, \\ 0 < u_1 < (1 - u_2)s_2/s_1 + u_2, \\ 0 < u_2, u_3 < 1 \end{cases} \end{aligned} \quad (15)$$

where $\mathbf{u} = [u_1, u_2, u_3]^T$ and γ is the SINR threshold.

B. Parameter Optimization via SPSA

The problem in (15) is non-convex. To solve this problem, it is relaxed as the following optimization problem

$$\begin{aligned} & \min_{\mathbf{u}} J = \text{PAPR} + \lambda \sum_{n=0}^{MN-1} \max(\gamma - \text{SINR}_n, 0)^2 \\ & \text{s.t.} \begin{cases} 0 < u_1 < (1 - u_2)s_2/s_1 + u_2 \\ 0 < u_2, u_3 < 1 \end{cases} \end{aligned} \quad (16)$$

where λ is a penalty coefficient, which makes a compromise between PAPR reduction and SINR loss in (16).

To solve the optimization problem in (16), the SPSA algorithm [13] is applied. In this algorithm, the parameter vector $\mathbf{u}$ is iteratively solved. In the i -th iteration, the gradient of the J is estimated by introducing the perturbation vector $\Delta^{(i)} = [\Delta_1^{(i)}, \Delta_2^{(i)}, \Delta_3^{(i)}]^T$, where the elements in $\Delta^{(i)}$ are independent with each other and follow the Bernoulli distribution with equal probability for $\Delta_k^{(i)} \in \{+1, -1\}$. The gradient of the k -th parameter of $\mathbf{u}$, denoted as $\hat{J}_k^{(i)}$, is approximately by the central difference

$$\hat{J}_k^{(i)} = \frac{J(\mathbf{u}^{(i)} + c^{(i)} \Delta^{(i)}) - J(\mathbf{u}^{(i)} - c^{(i)} \Delta^{(i)})}{2c^{(i)} \Delta_k^{(i)}}, k = 1, 2, 3 \quad (17)$$

where $c^{(i)}$ is the perturbation coefficient.

Based on the estimated gradient, k -th parameter of $\mathbf{u}$ is updated by

$$u_k^{(i+1)} = \mathcal{G} \left(u_k^{(i)} - a^{(i)} \hat{J}_k^{(i)} \right), k = 1, 2, 3 \quad (18)$$

where $a^{(i)}$ is the gain sequence, and $\mathcal{G}(\cdot)$ denotes the projection operator that ensures the updated parameters in (18) satisfy the constraints in (16). Specifically, $\mathcal{G}(\cdot)$ ensure that u_2 and u_3 are in $(0, 1)$ and u_1 is the interval $(0, (1 - u_2)s_2/s_1 + u_2)$.

The proposed piecewise companding method is summarized in **Algorithm 1**.

Algorithm 1: Piecewise Companding Method

Input: $I_{\max}, a_0, \eta, c_0, \zeta, \xi$

Output: Optimal companding parameters $\mathbf{u}_{\text{best}}$

- 1 Initialize: iteration index $i = 1$, $A = I_{\max}/10$,
 $\mathbf{u}^{(1)} = [u_1^{(1)}, u_2^{(1)}, u_3^{(1)}]^T$, $\mathbf{u}_{\text{best}} = \mathbf{u}^{(1)}$;
 - 2 **while** $i < I_{\max}$ **do**
 - 3 Update parameters: $a^{(i)} = \frac{a_0}{(i+A)^\eta}$, $c^{(i)} = \frac{c_0}{i^\zeta}$;
 - 4 Generate random perturbation vector $\Delta^{(i)}$;
 - 5 Calculate objective functions:
 $J_+^{(i)} = J(\mathbf{u}^{(i)} + c^{(i)} \Delta^{(i)})$,
 $J_-^{(i)} = J(\mathbf{u}^{(i)} - c^{(i)} \Delta^{(i)})$;
 - 6 Estimate gradient $\hat{J}_k^{(i)}, k = 1, 2, 3$ via (17);
 - 7 Update parameters $u_k^{(i+1)}, k = 1, 2, 3$ via (18);
 - 8 **if** $J(\mathbf{u}^{(i+1)}) \leq J(\mathbf{u}^{(i)})$ **then**
 - 9 $\mathbf{u}_{\text{best}} = \mathbf{u}^{(i+1)}$;
 - 10 **end**
 - 11 **if** $(J(\mathbf{u}^{(i+1)}) - J(\mathbf{u}^{(i)})) / J(\mathbf{u}^{(i+1)}) \leq \xi$ **then**
 - 12 **break**;
 - 13 **end**
 - 14 $i = i + 1$;
 - 15 **end**
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C. Computational Complexity

The computational complexity of the proposed piecewise companding method mainly determined by the SPSA-based

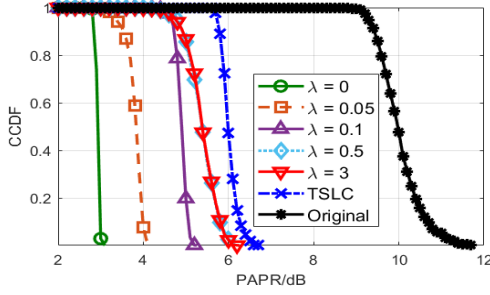


Fig. 1. CCDF of PAPR reduction under different penalty coefficients λ

parameter optimization. For an OTFS-ISAC block of length MN , the computational complexity of the objective function is $\mathcal{O}(MN)$. The computational complexity of the parameter update and projection steps is $\mathcal{O}(1)$ complexity. Therefore, the overall computational complexity of the proposed method is $\mathcal{O}(LMN)$, where L is the number of iterations.

IV. SIMULATION RESULTS

In this section, simulation results are presented to validate the effectiveness of the proposed piecewise companding method. Unless otherwise specified, the numbers of subcarriers M and symbols N are set as 128, the OTFS signal is modulated by QPSK, and the SINR threshold $\gamma = 15\text{dB}$. The rest of simulation parameters are given in Table I. In the following simulations, the proposed method is compared with the TSLC method and the original OTFS-ISAC signal.

TABLE I
SIMULATION PARAMETERS SETTING

Symbols	Parameters	Parameter values
$I_{\max}$	Maximum iterations	60
a_0	Initial gain coefficient	0.2
η	Gain decay rate	0.6
c_0	Initial perturbation coefficient	0.1
ζ	Perturbation decay rate	0.1
ξ	Convergence threshold	1×10^{-6}

Fig. 1 illustrates the complementary cumulative distribution function (CCDF) of PAPR of OTFS-ISAC. It can be seen that the PAPR is decreased with the decrease of λ . When $\lambda > 0.5$, the PAPR is almost unchanged. Moreover, the proposed method is always has the best PAPR performance compared with the TSLC method and the original OTFS-ISAC signal.

Fig. 2 shows the variation of BER with SNR. We can see that the performance of OTFS-ISAC signal in terms of BER is improved with the increase of λ . When $\lambda > 0.1$, the proposed method is better than TSLC method. Moreover, When $\lambda > 0.5$, the proposed method can almost achieve the same BER as the original OTFS-ISAC signal. From Fig. 1 and Fig. 2, we can see that the PAPR and BER performance can be adjusted by λ , and we have to make a trade-off between PAPR and BER.

To evaluate the performance of the proposed method under different number of subcarriers M , the CCDF of PAPR and the

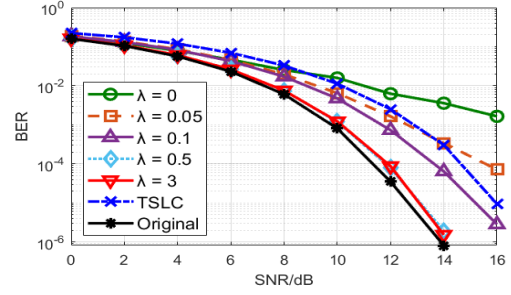
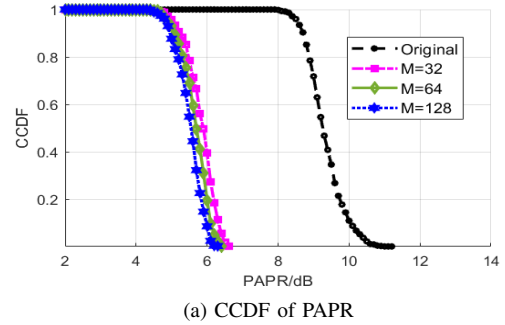
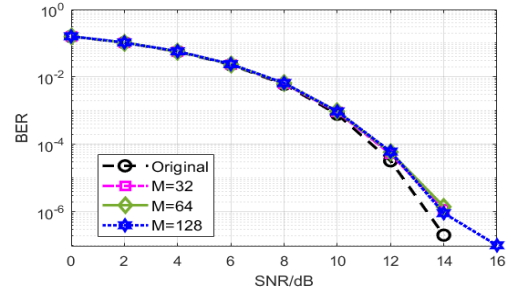


Fig. 2. BER versus SNR for different penalty coefficients λ

BER versus SNR are given in Fig. 3. In Fig. 3, $N = 64$ and $\lambda = 0.5$. It can be seen that the proposed method achieves significant PAPR reduction, and the PAPR performance is improved with increasing number of subcarriers. In addition, with the increase of the number of subcarriers, the BER is almost the unchanged.



(a) CCDF of PAPR



(b) BER versus SNR

Fig. 3. PAPR and BER performance under different numbers of subcarriers with fixed $N = 64$ and $\lambda = 0.5$.

V. CONCLUSION

In this paper, we proposes a piecewise companding method for OTFS-ISAC signal to reduce PAPR. In this method, the amplitude of the OTFS-ISAC signal is divided into five parts, and the relationship between the companding parameters and the SINR is derived by using the Busgang theorem. Moreover, the companding parameters are optimized by the SPSS algorithm. Simulation results demonstrate that the proposed method significantly reduces PAPR while maintains BER performance. Compared with the TSLC method, the proposed method can achieve a better performance.

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